

Exercise and chronic kidney disease: potential mechanisms underlying the physiological benefits

Nicolette C. Bishop^{1,2} , James O. Burton^{1,2,3,4}, Matthew P. M. Graham-Brown^{2,3,4}, David J. Stensel^{1,2,5,6}, João L. Viana⁷ & Emma L. Watson^{2,3}

Abstract

Increasing evidence indicates that exercise has beneficial effects on chronic inflammation, cardiorespiratory function, muscle and bone strength and metabolic markers in adults with chronic kidney disease (CKD), kidney failure or kidney transplants. However, the mechanisms that underlie these benefits have received little attention, and the available clinical evidence is mainly from small, short-duration (<12 weeks) exercise intervention studies. The available data, mainly from patients with CKD or on dialysis, suggest that exercise-mediated shifts towards a less inflammatory immune cell profile, enhanced activity of the NRF2 pathway and reduced monocyte infiltration into adipose tissue may underlie improvements in inflammatory biomarkers. Exercise-mediated increases in nitric oxide release and bioavailability, reduced angiotensin II accumulation in the heart, left ventricular remodelling and reductions in myocardial fibrosis may contribute to improvements in left ventricular hypertrophy. Exercise stimulates an anabolic response in skeletal muscle in CKD, but increases in mitochondrial mass and satellite cell activation seem to be impaired in this population. Exercise-mediated activation of the canonical wnt pathway may lead to bone formation and improvements in the levels of the bone-derived hormones klotho and fibroblast growth factor 23 (FGF23). Longer duration studies with larger sample sizes are needed to confirm these mechanisms in CKD, kidney failure and kidney transplant populations and provide evidence for targeted exercise interventions.

Sections

[Introduction](#)[Chronic inflammation](#)[Cardiovascular health](#)[Lipid metabolism and insulin resistance](#)[Sarcopenia](#)[Bone health](#)[Conclusions](#)

Key points

- Shifts towards a less inflammatory monocyte and T cell profile, enhanced activity of the NRF2 pathway and reduced monocyte infiltration into adipose tissue might underlie exercise-mediated improvements in inflammatory biomarkers in people with chronic kidney disease (CKD) or on dialysis.
- Exercise-mediated enhancements in nitric oxide release and bioavailability, reduced angiotensin II accumulation in the heart, left ventricular remodelling and reductions in myocardial fibrosis might improve left ventricular hypertrophy in CKD and kidney failure.
- Few studies have examined the effects of exercise training on insulin resistance in people with CKD; however, early evidence suggests that a 6-month exercise intervention might have beneficial effects.
- Exercise, particularly resistance exercise, stimulates an anabolic response in skeletal muscle in people with CKD; exercise-induced increases in mitochondrial mass and satellite cell activation seem to be impaired in this population.
- Exercise-mediated activation of the canonical wnt pathway may lead to bone formation in CKD; reduced fibroblast growth factor-23 (FGF23) release and increased klotho release in bone might also contribute to improved bone mineral density and reduced bone loss.
- Studies with longer durations and larger sample sizes are needed to confirm these potential mechanisms in all CKD and kidney failure populations and provide evidence to enable the development of effective targeted exercise interventions.

Introduction

A substantial body of evidence indicates that exercise – defined as physical activity involving planned, structured and repetitive bodily movement¹ – enhances physical function, physical capacity and quality of life in people with kidney disease² and might reduce the risk of common comorbidities, such as cardiovascular disease (CVD), obesity and diabetes in this population. Such evidence contributed to the development of the Renal Association clinical practice guideline for exercise and lifestyle in chronic kidney disease (CKD)² that summarizes the evidence for the benefits of regular exercise participation for people with all stages of kidney disease.

Although the number of studies describing how exercise can regulate inflammation, cardiovascular health, lipid metabolism and insulin resistance, sarcopenia and bone health in CKD is growing, far less research has focused on understanding the potential mechanisms underlying these benefits. Such knowledge is necessary to enable the development of targeted interventions to optimize the benefits of physical activity and exercise alone and in conjunction with existing treatments. In the future habitual physical activity and prescribed exercise interventions could act as adjuvants to existing therapies to increase their effectiveness. Understanding the mechanisms that underlie the beneficial effects of exercise might also lead to the identification of novel therapeutic targets and key areas for future research.

In this exploratory Review, we discuss the potential mechanisms by which regular exercise participation has beneficial effects on chronic

inflammation, cardiovascular health, lipid metabolism, insulin resistance, sarcopenia and bone health in people with CKD, kidney failure or kidney transplants. We also highlight gaps in the evidence and areas for priority investigation to further enhance the benefits of exercise for those living with kidney disease.

Chronic inflammation

Uraemia is associated with dysfunction of almost every aspect of the immune system³, which manifests as immunosuppression, reduced responsiveness to antigen challenge (including vaccination)^{3,4} and chronic activation of immune cells leading to persistent release of pro-inflammatory cytokines and chronic systemic inflammation^{5,6}. Immune dysfunction and chronic inflammation progress with loss of kidney function^{7,8} and are key contributors to the greatly increased risk of CVD and infections, two major causes of mortality, among people with CKD⁹. Immune cell over-activation and chronic inflammation also contribute to increased risk of developing insulin resistance, type 2 diabetes mellitus (T2DM), anaemia, sarcopenia and osteoporosis in these patients^{4,5}. Circulating levels of the pro-inflammatory cytokine IL-6 are reported to be the strongest predictors of comorbidity and outcome in people with CKD¹⁰.

Compared with the volume of studies reporting exercise-mediated alterations in circulating pro-inflammatory (e.g. IL-6 and tumour necrosis factor (TNF)) and anti-inflammatory (e.g. IL-10) cytokines in CKD^{11–13}, investigations of the mechanisms that underlie these anti-inflammatory effects across all stages of kidney disease are limited. Many mechanisms by which regular exercise confers anti-inflammatory effects in other populations with chronic inflammation, such as people with CVD, T2DM, rheumatoid arthritis or non-alcoholic fatty liver disease, require confirmation in patients with CKD. However, clinical data suggest that regular exercise can alter the inflammatory characteristics of circulating immune cells in these patients. A study of 18 patients reported that a 6-month programme of brisk walking for 30 min, 5 times per week resulted in reductions in the numbers of CD4⁺ and CD8⁺ T cells that expressed the activation marker CD69 following stimulation with the super-antigen staphylococcus enterotoxin B (SEB)¹¹. The numbers of these cells did not change over the 6-month study period in 14 patients who received usual care. Activation of SEB-stimulated monocytes (indicated by upregulation of CD86 and HLA-DR) was reduced in the walking group but elevated in the usual care group after 6 months. This dampening of immune cell activation likely contributed to a decrease in the circulating inflammatory status of the walking group; IL-6 tended to decrease whereas IL-10 increased in this group, resulting in a reduction in the IL-6 to IL-10 ratio. No change in this ratio was seen in the usual care group. The levels of the soluble type I and type II TNF receptors also showed a tendency to decrease in the walking group only; these receptors bind to and sequester TNF, limiting its inflammatory actions¹⁴. Plasma concentrations of C-reactive protein (CRP) did not change in response to the walking intervention, suggesting that CRP release is not dependent on and therefore does not necessarily reflect changes in IL-6 release.

Clinical data also support an exercise-associated shift towards an anti-inflammatory circulating immune cell profile in patients on haemodialysis. A study that included 22 patients reported a decrease in the proportion of highly pro-inflammatory intermediate monocytes in those who participated in a 6-month, thrice weekly intradialytic cycling programme compared with no change in 16 non-exercising patients who received usual care¹⁵. Intermediate monocytes secrete pro-inflammatory TNF, IL-1 β and IL-6, express the chemokine receptors

CCR5 and CX3CR1, which enhance their migratory potential towards tissue sites of inflammation¹⁶, and are associated with increased CVD risk in patients with CKD and in patients on haemodialysis^{17,18}. The usual care group also showed a decrease from baseline in numbers of regulatory T (T_{reg}) cells, which inhibit T cell-derived inflammation and release IL-10, whereas T_{reg} cell numbers increased from baseline in the intradialytic exercise group. This beneficial effect on T_{reg} cells might contribute to the anti-inflammatory effects of exercise in these patients.

Regular exercise also has a beneficial effect on circulating markers of chronic inflammation in kidney transplant recipients (KTRs)¹⁹. However, a study in 26 KTRs reported that lymphocyte and monocyte population phenotypes did not change in response to 2 months of regular moderate or high-intensity interval training²⁰. This finding may indicate that exercise interventions need to be performed for at least 3 months to obtain an anti-inflammatory effect. The immunosuppressive medication that is taken by KTRs might also reduce the potential for exercise-induced changes in immune cell phenotypes. However, the absence of a shift towards a more pro-inflammatory monocyte and T cell profile suggests that moderate- and high-intensity interval exercise is safe for KTRs from an immune standpoint.

Alterations in circulating immune cell populations might partly account for the observation that a programme of intradialytic resistance exercise three times per week for 3 months increased peripheral blood mononuclear cell (PBMC) nuclear factor erythroid-2 related factor 2 (NRF2) mRNA expression compared with pre-intervention levels in 25 patients²¹. No change in PBMC NRF2 mRNA expression was observed in 19 patients on haemodialysis who received usual care. This observation is promising given that people with CKD or kidney failure typically have reduced activity of the NRF2 pathway²², which has an important role in impeding NF- κ B signalling pathways that are central to de novo synthesis of many inflammatory cytokines²³. Unfortunately, the effect of the intradialytic exercise intervention on pro-inflammatory cytokine gene expression in PBMCs was not assessed in this study. However, the results provide intriguing early support for exercise-enhanced NRF2 pathway activation as a potential mechanism for the anti-inflammatory effects of exercise in patients receiving haemodialysis, which should also be investigated across the spectrum of CKD and kidney failure. Such enhanced activation could occur via preferential mobilization of cells with high activity of the NRF2 pathway and/or via increased activation of the NRF2 pathway in circulating immune cells.

Migration of bone marrow-derived monocytes as infiltrating macrophages is a crucial contributor to the generation of adipose tissue inflammation, which is the major source of circulating IL-6 at rest. Such migration might be reduced by regular exercise participation²⁴. Patients with kidney failure had greater macrophage infiltration of subcutaneous and visceral adipose tissue than healthy individuals, independent of adipose tissue mass²⁵. The effect of exercise interventions on monocyte migration and macrophage infiltration has not been determined in CKD. However, a small study in 8 patients reported that a 20-min session of moderate intensity intradialytic exercise attenuated the elevations in proportions of CCR5-expressing CD4⁺ T cells and CD14⁺ monocytes that were observed during a normal haemodialysis session²⁶. As CCR5 expression enhances monocyte migration towards inflamed tissue, these observations provide cautious early support for the hypothesis that an exercise-mediated reduction in adipose tissue inflammation might contribute to the anti-inflammatory effects of exercise in CKD.

Uraemia is associated with accelerated immunosenescence of T cells and monocytes in people with kidney failure²⁷. Immunosenescence

is both a consequence of and a contributor to immune dysfunction, increased risk of infections and the development of chronic inflammation²⁸. Habitual physical activity is associated with slowing of the progression of immune system ageing in older adults²⁸, suggesting that investigation of the effect of physical activity on premature immune ageing in CKD could be an impactful avenue for future research.

In summary, uraemia is associated with immune dysfunction that manifests as both immune suppression and sustained cellular immune overactivation leading to the development of chronic inflammation. Increasing levels of habitual physical exercise reduced the levels of markers of chronic inflammation in patients with CKD, patients on dialysis and KTRs in many, but not all, studies. Exercise-mediated phenotypic alterations in monocyte and T cell subpopulations that promote an anti-inflammatory leukocyte profile have been shown in non-dialysis and haemodialysis populations but not in KTRs (Fig. 1). These alterations might reflect preferential mobilization of subpopulations of cells with a less inflammatory phenotype and/or altered expression of cell surface receptors associated with inflammatory functions (e.g. CD69, CCR5). Early evidence suggests that regular intradialytic exercise results in increased activity of the NRF2 pathway in patients on haemodialysis (Fig. 1). This finding could reflect the mobilization of cells with higher NRF2 activity or direct stimulation of the NRF2 signalling pathway by an unknown exercise-related mediator. The role of regular exercise participation in regulating uraemia-mediated immune senescence in people with CKD is an intriguing avenue for future research.

Cardiovascular health

In people with CKD, the risk of developing CVD is far higher than the risk of kidney failure, making CKD one of the strongest risk factors for the development of CVD²⁹. Meta-analyses of cohort studies involving >1.3 million people with CKD who were not on dialysis clearly demonstrate that loss of kidney function is associated with CVD and mortality, even after adjustment for other established risk factors^{30,31}. Among patients with CKD stage 4–5, half of them have CVD³² and cardiovascular mortality accounts for around 40–50% of all deaths, compared with 26% of deaths in people with normal kidney function^{33,34}. The risk of CVD among patients on haemodialysis is even higher; >60% of these patients have CVD³⁵ and their cardiovascular mortality is estimated to be between 10- and 100-fold higher than that of the general population³⁶.

Physical fitness (usually quantified as peak or maximum oxygen uptake or a surrogate marker of this uptake) is an independent predictor of CVD and all-cause mortality. A systematic review and meta-analysis of six studies that included 512 participants with CKD who were not on dialysis observed that peak oxygen uptake was lower in those with CKD stages 3b–5 than in those with CKD stages 2–3a³⁷. This finding suggests that physical fitness declines as CKD progresses. People with CKD are therefore at an increased risk of CVD owing to a direct (independent) association between low fitness and CVD risk as well as the adverse effects of reduced physical fitness on CVD risk factors including blood lipids and insulin resistance. Declines in physical fitness are associated with declines in physical activity that exacerbate CVD risk. A sedentary lifestyle has been shown to worsen cardiovascular outcomes and mortality in dialysis-dependent³⁸ and non-dialysis-dependent patients with CKD^{39,40}, with corresponding improvements in outcomes observed in those individuals who are more physically active.

Traditional cardiovascular risk factors such as hypertension, diabetes and dyslipidaemia are highly prevalent in people with CKD and

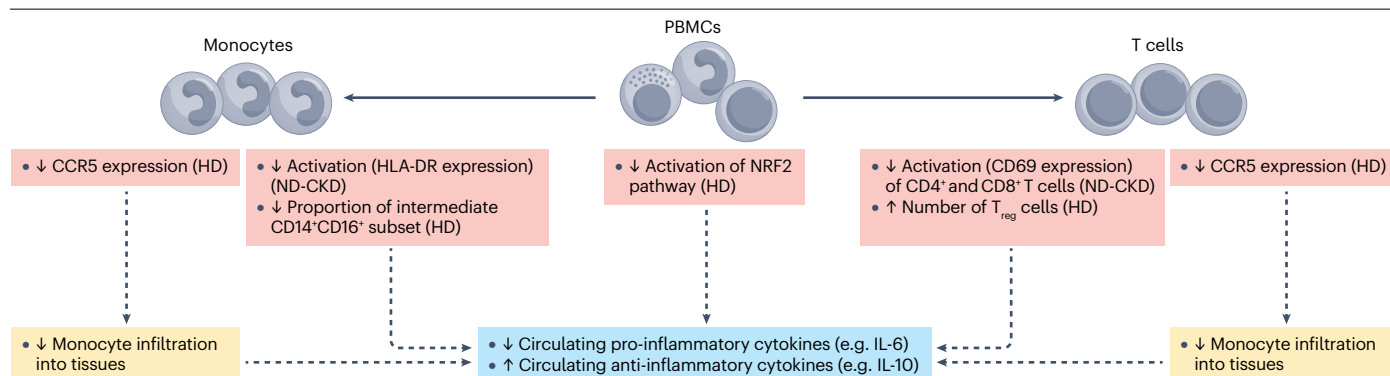


Fig. 1 | Potential mechanisms that could underlie the anti-inflammatory effects of exercise in non-dialysis chronic kidney disease and haemodialysis populations. Intradialytic exercise may enhance the activity of the nuclear factor erythroid-2 related factor 2 (NRF2) pathway in peripheral blood mononuclear cells (PBMCs)²¹. This pathway impedes nuclear-factor Kappa B signalling and subsequent pro-inflammatory cytokine synthesis. A study in patients with non-dialysis chronic kidney disease (ND-CKD), reported that 6 months of regular walking exercise resulted in a reduction in the activation of monocytes (determined by HLA-DR expression) and T cells (determined by CD69 expression) following stimulation of PBMCs with staphylococcus enterotoxin B (SEB)¹¹. In patients on haemodialysis, 6 months of intradialytic exercise led to a reduction

in the proportion of the pro-inflammatory intermediate monocyte subset and an increase in the numbers of regulatory T (T_{reg}) cells, compared with patients who received usual care¹⁵. These phenotypic changes also likely contribute towards reported exercise-mediated reductions in circulating pro-inflammatory cytokines (e.g. IL-6) and increases in anti-inflammatory cytokines (e.g. IL-10). Intradialytic exercise has also been associated with reduced monocyte and T cell expression of the chemokine receptor CCR5, which is known to be involved in the process of immune cell infiltration into tissue sites, including adipose, cardiac tissue, muscle and vasculature. This infiltration is a key event in the release of pro-inflammatory cytokines into the circulation.

make a substantial contribution to the development of CVD in those with early-stage CKD. However, CKD is also an independent risk factor for CVD^{41,42}. Cardiovascular survival improves after kidney transplantation, even in high-risk patients⁴³, suggesting a direct role of uraemic factors in the pathogenesis of CVD in CKD. This finding highlights the importance of non-traditional risk factors, such as left ventricular hypertrophy (LVH), which is one of the hallmarks of uraemic cardiomyopathy. LVH is present in about one-third of people with CKD and 75% of those with kidney failure, and is almost ubiquitous in patients who have been on dialysis for >2 years⁴⁴. The presence of LVH is an independent predictor of mortality across the CKD spectrum and interventions that induce reductions in left ventricular mass (e.g. exercise programmes, extended dialysis regimens and correction of anaemia) are associated with significant reductions in the risk of all-cause mortality in patients with CKD⁴⁵. The causes of LVH in CKD are multi-factorial and include factors related to afterload, factors related to preload and non-haemodynamic factors⁴⁶. Increasing evidence suggests that regular exercise can influence these mechanisms in people with CKD.

Afterload-related factors include abnormal arterial stiffness, increased systemic arterial resistance and systolic hypertension. Regular participation in aerobic exercise may decrease pulse wave velocity in healthy individuals and in people with CKD^{47,48}. This potential improvement in arterial stiffness in response to exercise training has been proposed to result from improved endothelial function, resulting in an increase in the release of nitric oxide and a consequent improvement in vasodilatation^{47,49}. Enhanced bioavailability of nitric oxide is also speculated to be associated with improvements in arterial compliance in people with CKD⁵⁰. These factors contribute to the reductions in systolic and diastolic blood pressure that are reported with exercise training more globally⁵¹.

The term 'pre-load factors' almost exclusively relates to volume status. People with CKD, including those on dialysis, are prone to salt and water retention and volume overload, which increases their cardiac

pre-load. Although exercise might not directly mitigate preload factors, optimizing fluid status in patients on dialysis and in those with non-dialysis-dependent CKD is recommended to derive the maximal benefit from exercise programmes.

Regular exercise programmes might also influence non-afterload-, non-preload-related factors including intracellular mediators and pathways that contribute to progressive LVH. In people with CKD who were not on dialysis, a resistance exercise training programme resulted in improvements in chronic inflammation and biomarkers of anaemia⁵². These beneficial effects might also reduce the preload factors that contribute to LVH. Some evidence from single centre studies suggests that intradialytic exercise can reduce hyperphosphataemia, which is associated with hypertension and increased left ventricular mass^{53,54}. However, this reduction in phosphate levels might be an indirect effect related to increased clearance of this highly dialyzable molecule owing to the increase in rate of blood flow that occurs with exercise. Studies in animal models of kidney failure suggest that regular exercise abrogates excess angiotensin II accumulation in the heart, which promotes myocyte hypertrophy and interstitial fibrosis⁵⁵.

In the CYCLE-HD trial, a 6-month programme of regular intradialytic exercise resulted in a clinically relevant decrease in left ventricular mass that was associated with beneficial left ventricular remodelling⁵⁶. The exercise group ($n = 51$) showed significant reductions in myocardial native T1 times (measured using cardiac MRI), suggesting reductions in myocardial inflammation and fibrosis. These factors directly contribute to the development of cardiac arrhythmias and sudden cardiac death and patients with CKD have a disproportionately high risk of these cardiovascular events. Native myocardial T1 times increase incrementally with CKD stage⁵⁷ and regular aerobic exercise has been reported to induce favourable left ventricular remodelling in adults with hypertension⁵⁶ suggesting that exercise interventions might be a promising non-pharmacological approach for the prevention of sudden cardiac death and arrhythmias in CKD.

Several potential mechanisms could have driven the adaptations reported in the CYCLE-HD trial⁵⁶, including improvements in arterial compliance or reductions in cardiac stunning and chronic systemic inflammation^{58,59}. The patients cycled at very low intensities, even lower than those attained by patients undergoing cardiac rehabilitation⁶⁰, suggesting that the exercise intensity was too low for the observed benefits to have been conferred through traditional mechanisms, such as improved cardiorespiratory thresholds.

An alternative hypothesis is that microRNAs (miRs) released from contracting muscle, even during low-intensity exercise, might have a role in beneficial left ventricular remodelling. MiRs have important roles in the pathological processes of CVD^{61,62} and could potentially also be involved in conferring the cardioprotective effects of exercise. Skeletal muscle can synthesize and secrete hundreds of proteins and extracellular vesicles into the circulation in response to muscular contraction⁶³. Exosomes are small endogenous membrane-bound extracellular vesicles that can facilitate cell–cell communication via the transmission of a variety of signalling molecules, including miRs⁶⁴. The release of exosomes into the circulation following exercise⁶⁵ has emerged as a potential mechanism through which the beneficial effects of exercise are transmitted to distal tissues and organs, including the heart⁶⁶. For example, exercise modulates the expression of circulating miRs, which may lead to attenuation of cardiac hypertrophy, protection against abnormal ventricular remodelling and even confer cardioprotection⁶⁷.

Little is known about the effects of exercise on the exosome miR profile in the non-dialysis CKD population. A small study in these patients reported that an acute bout of cycling resulted in an increase in circulating miR-150 (which is known to be cardioprotective) and reductions in miRs-210 and -146a (which are both associated with heart failure)⁶⁸. In this study, 12 weeks of home-based aerobic training resulted in a reduction in the upregulation of circulating miR-210 that occurred in response to a single exercise session. The long-term effects of exercise on beneficial miR expression in patients with CKD are unknown.

In summary, the mechanisms that underlie the disproportionately high burden of CVD in people with CKD are multifactorial. However, clear evidence indicates that interventions aimed at increasing levels of physical activity can have a positive impact across the spectrum of CKD. These findings highlight the importance of implementing exercise programmes that are recommended in clinical guidelines for these patients². Further research is needed to identify novel interventions that mimic the cardioprotective benefits of exercise in CKD.

Lipid metabolism and insulin resistance

Dyslipidaemia and insulin resistance are hallmarks of CKD. People with CKD often have hypertriglyceridaemia, elevated levels of low-density lipoprotein cholesterol (LDL-C) and atherogenic small, dense LDL particles and low levels of high-density lipoprotein cholesterol (HDL-C) owing to impaired metabolism of triglyceride-rich lipoproteins, including very-low density lipoprotein, intermediate density lipoprotein and chylomicrons. This metabolic impairment may be related to reductions in the activity of lipoprotein lipase (LPL), which hydrolyses triglyceride from triglyceride-rich lipoproteins, and lecithin-cholesterol acyltransferase (LCAT), which has a critical role in the synthesis of new HDL-C particles. The decline in HDL-C owing to reduced LCAT activity is exacerbated by reduced circulating concentrations of apolipoprotein A-I, which is a structural protein essential for HDL-C formation. These alterations reduce the capacity for removal of excess

cholesterol via reverse cholesterol transport⁶⁹. Notably, a lipid paradox exists in the advanced stages of CKD whereby low serum cholesterol levels are associated with higher mortality. This paradox might be explained by high levels of systemic inflammation and oxidative stress, which increase the levels of oxidized LDL-C⁷⁰.

Insulin resistance is an early feature of CKD and affects almost all people with kidney failure⁷¹. Numerous mechanisms linking insulin resistance and CKD have been proposed, including glucose dysregulation, uraemic toxins, hypertension, chronic systemic inflammation, oxidative stress, abnormal mineral metabolism, adiposity and lipotoxicity^{72–74}. Muscle is the primary site for insulin resistance and CKD inhibits insulin signalling pathways within muscle via multiple mechanisms⁷⁵. Some studies suggest that insulin resistance is an independent risk marker for CVD in people with CKD⁷⁶, but the data are inconsistent. For example, no association was observed between Homeostasis Model Assessment of Insulin Resistance (HOMA-IR) and CKD progression, CVD risk or all-cause mortality in the Chronic Renal Insufficiency Cohort Study⁷⁷.

Cardiorespiratory fitness (that is, maximum oxygen uptake) is notably lower in people with CKD than in healthy individuals^{78,79}, which may be a cause and consequence of low physical activity levels in people with CKD. A cross-sectional study involving 47 people with CKD (estimated GFR <60 ml/min/1.73 m²) and 29 healthy individuals used accelerometry to determine physical activity levels and a hyperinsulinaemic–euglycaemic clamp to assess insulin sensitivity. The results suggested that participants with CKD were less active and had lower insulin sensitivity and higher serum triglyceride than healthy individuals. Physical activity was similarly associated with insulin sensitivity and serum triglyceride in participants with and without CKD⁸⁰.

Early studies demonstrated that people on haemodialysis can exercise at a moderate intensity (40–50% of maximum oxygen uptake) for prolonged periods without adverse cardiovascular or metabolic responses^{78,79} and that endurance exercise training (cycling, walking and jogging) has the potential to beneficially modify CVD risk factors (e.g. triglyceride and HDL-C) in these patients⁸¹. Several systematic reviews have examined the effects of exercise training on CVD risk markers in people with CKD^{82,83}. An analysis of 59 randomized controlled trials (RCTs) across all stages of kidney disease (9 in patients not on dialysis, 47 in patients on haemodialysis and 3 in KTRs) found

Glossary

Accelerometry

An objective measurement of movement (acceleration) that is used to quantify the volume and intensity of physical activity.

Blood flow restriction training

A technique that combines low-intensity exercise with blood flow occlusion through the application of a cuff or band with the aim of producing comparable effects with high-intensity exercise.

Dynamic resistance exercise

Effort against a constant force that involves movement of a joint through

the range of motion, for example, lifting weights or press ups.

Isometric resistance exercise

Static contractions of a specific muscle or group of muscles with no resultant change in muscle length or joint movement.

Mechanotransduction

The conversion of mechanical loading into biochemical signalling.

that most had small sample sizes and a high risk of bias⁸². Moreover, most of the trials that assessed lipid profiles found no effect of exercise on LDL-C, HDL-C or triglycerides. Similarly, a systematic review and meta-analysis including 31 trials that enrolled a total of 1,305 adults with CKD⁸³ found no beneficial effect of aerobic exercise interventions on LDL-C, total cholesterol or triglyceride levels. However, a subset analysis of six trials with 155 participants that examined HDL-C levels identified increased concentrations after the exercise intervention. Another systematic review that identified 15 trials that examined cardiometabolic and renal outcomes in 622 people with CKD concluded that aerobic exercise interventions had no beneficial effects on blood lipids or HbA1c levels⁸⁴. These findings suggest that aerobic exercise interventions are ineffective for altering most lipids and lipoproteins. Although some studies suggest that exercise might elevate HDL-C levels in people with CKD⁸⁵, others reported no change in HDL-C concentration or particle size following an exercise intervention⁸⁶. The results of an RCT published in 2020 suggest that 12 months of balance and strength training might decrease levels of atherogenic lipoprotein (a) in patients with CKD, but these findings require confirmation in further studies⁸⁷.

Regular exercise has the potential to combat insulin resistance in people with CKD by enhancing insulin signalling in skeletal muscle⁷⁵. Unfortunately, few studies have investigated the effects of exercise on insulin resistance in this population and the available evidence is inconsistent. A small study involving 11 patients on haemodialysis who performed exercise on treadmills and/or recumbent bikes over a 3-month period found no impact of the intervention on HOMA-IR⁴⁸. Another small study in which 9 participants with CKD and obesity were enrolled on a 12-week lifestyle intervention programme involving supervised exercise and dietary counselling observed improved insulin sensitivity indicated by the Matsuda Index (determined from an oral glucose tolerance test) but no significant change in insulin resistance assessed using HOMA-IR⁸⁸. By contrast, a 24-week dynamic resistance training programme improved glucose tolerance (assessed using an oral glucose tolerance test), lowered HOMA-IR and reduced HbA1c levels in participants undergoing haemodialysis⁸⁹. This result is consistent with the findings of a study in people with impaired glucose tolerance or T2DM undergoing haemodialysis that demonstrated that strength training (three times per week over 16 weeks) significantly improved glucose tolerance (assessed using an oral glucose tolerance test)⁹⁰. These data suggest that interventions of longer duration that involve resistance and/or strength training are more likely than shorter interventions to elicit improvements in glucose tolerance and/or insulin sensitivity in people living with CKD and kidney failure.

New onset diabetes after transplantation (NODAT) is a frequent complication of kidney transplantation and the effects of exercise training on insulin resistance, blood lipids and other CVD risk factors in KTRs constitute an area of increasing research interest. Systematic reviews have reported limited evidence of the beneficial effects of exercise interventions on levels of lipids and lipoproteins or on glucose tolerance and insulin resistance in this population^{91,92}. However, these outcomes are rarely, if ever, the primary outcomes of studies in KTRs and the available studies might have been underpowered in this respect.

The potential of exercise to improve blood lipids and insulin resistance in KTRs might depend on the extent of dyslipidaemia and/or glucose intolerance. A RCT in 21 KTRs with T2DM reported that a 6-month programme of aerobic and muscle-strengthening exercises resulted in reductions in fasting plasma glucose, triglycerides

and HbA1c, and increases in HDL-C⁹³. Few studies have used the hyperinsulinaemic–euglycaemic clamp to assess insulin sensitivity in KTRs. However, a study that used this technique reported no significant difference in glucose disposal rate before and after kidney transplantation in 5 participants (4 men and 1 woman)⁹⁴. This study also reported that kidney transplantation did not result in an improvement in skeletal muscle gene expression following an acute bout of exercise; such expression is reduced in people with CKD⁹⁴. A pilot study in 24 KTRs who were randomly assigned to a 16-week resistance training intervention or a control group showed that resistance training is feasible and improves muscle strength in these patients; however, no effect of the intervention on the incidence of NODAT was observed⁹⁵. Thus, the available data do not suggest a benefit of exercise for NODAT⁹⁶ or glycaemic control in KTRs⁹².

In summary, limited evidence is available to support a beneficial effect of exercise on lipid and lipoprotein metabolism or insulin resistance in people with CKD. This failure might be partly due to study limitations, including small sample sizes and short duration of interventions. Most of the published studies are in people undergoing haemodialysis and few have focused on patients with the earlier stages of CKD or KTRs. No data are available from studies examining the effects of exercise on enzymes involved in lipid and lipoprotein metabolism (i.e., LPL and LCAT) in people with CKD and few studies have employed the hyperinsulinaemic–euglycaemic clamp to assess insulin sensitivity in participants with CKD. Researchers have postulated that exercise as a monotherapy might be insufficient for altering CVD risk factors in people with CKD and administration of medications such as steroids and calcineurin inhibitors might counteract the potential benefit of exercise in KTRs⁹¹. The extent to which exercise reduces the need for glucose- and lipid-lowering medication in people with CKD and KTRs is an area for future research.

Sarcopenia

Loss of muscle strength, mass and function, termed sarcopenia, is common in CKD⁹⁷. Sarcopenia begins early in the disease process⁹⁸ and leads to progressive muscle wasting, muscle disuse, a reduction in strength and poor outcome. Muscle wasting is primarily driven by an increased rate of protein degradation via the ubiquitin proteasome system and an increase in the catabolic protein myostatin, which suppresses the growth of skeletal muscles and increases atrophy. Metabolic acidosis, uraemic toxins, insulin resistance, inflammation, dysregulated miR expression and physical inactivity contribute to muscle wasting in CKD⁹⁷. Skeletal muscle biopsy samples from people on haemodialysis showed a reduction in fibre size cross-sectional area, with type II fibres more severely affected than type I fibres⁹⁹ and a reduction in the percentage of type I fibres¹⁰⁰, suggesting that CKD affects fibre size and composition. At the cellular level, such samples show evidence of intramuscular inflammation¹⁰¹, low insulin-like growth factor-1 (IGF-1) concentrations¹⁰², fibre splitting and abnormal fibre type grouping¹⁰³. Skeletal muscle cells from people with CKD also show reduced mitochondria number¹⁰⁴ and mitochondria that appear to be swollen with absent cristae⁹⁹, resulting in impaired mitochondrial energy metabolism¹⁰⁵. Importantly, skeletal muscle is highly adaptive and easily remodelled through interventions such as exercise. The available data clearly demonstrate a beneficial effect of exercise on skeletal muscle in CKD; however, the molecular responses to exercise in this population remain to be fully elucidated.

Resistance exercise training is generally associated with an increase in skeletal muscle mass, including in people with CKD^{106,107}.

A study in patients with CKD who were not on dialysis (10 in the exercise group and 7 in the control group) showed large increases in the skeletal muscle mRNA expression of two muscle-specific E3 ligases, MuRF-1 and MAFbx, as well as increases in IL-6, monocyte chemoattractant protein-1 and TNF expression 24 h after a session of acute resistance exercise¹⁰⁸. These catabolic and inflammatory changes are a normal response to unaccustomed exercise that is required to remove damaged muscle tissue. Following 8 weeks of resistance exercise training, this skeletal muscle response was no longer observed 24 h after an exercise session, demonstrating that the patients had likely become accustomed to the exercise. In adults without CKD, phosphorylation of Akt, a key protein in the insulin signalling pathway, is reported to increase following unaccustomed resistance exercise and remain high for up to 48 h¹⁰⁹. However, an acute resistance exercise session did not result in an increase in Akt phosphorylation in people with CKD¹⁰⁸. Critically this Akt response to exercise was restored following the 8-week resistance training intervention. The lack of an initial Akt response in people with CKD might indicate impairments in the IGF-1–Akt signalling pathway in response to unaccustomed exercise, which seem to be overcome by training. These alterations in molecular responses to exercise following resistance exercise training likely contribute to exercise-induced increases in muscle mass.

In 51 patients on haemodialysis, 18 weeks of strength training, aerobic training or combined strength and aerobic training significantly reduced the mRNA expression of myostatin, and increased mRNA expression of the IGF-1 splice variants IGF-1Ea, IGF-1Eb and IGF-1Ec¹¹⁰. IGF-1 has an important anabolic role in skeletal muscle, increasing rates of protein synthesis and reducing protein degradation, resulting in hypertrophy. Interestingly, the increase in IGF-1 mRNA was seen in the absence of corresponding increases in IGF-1 protein concentrations, which the researchers suggested might be explained by biopsy timings that captured transcriptomic but not proteomic changes. In another study by the same group, 8 weeks of aerobic exercise training in 10 sedentary patients on haemodialysis also resulted in a significant reduction in myostatin mRNA expression and increases in IGF-1 receptor and the IGF binding proteins IGFBP2, IGFBP4 and IGFBP5 (ref. ¹¹¹) but no change in fibre size. These results demonstrate that in patients with CKD, including those on dialysis, exercise can activate anabolic pathways that can result in skeletal muscle hypertrophy if the overload signal is sufficient. However, another study in 40 patients with CKD not on dialysis reported no effect of 6 months of home-based walking (aerobic) exercise on anabolic or catabolic processes in skeletal muscle¹¹², with no changes in mRNA expression of MuRF-1 or MAFbx, basal phosphorylation of Akt or muscle mass¹¹³. Interestingly, the participants showed a large depletion of intramuscular amino acids that could be prevented by supplementation with sodium bicarbonate. This supplementation in combination with walking exercise significantly reduced MuRF-1 mRNA expression but had no effect on IGF-1–Akt signalling¹¹². The timings of the muscle biopsies prevent firm conclusions from being drawn, but it is likely that walking exercise did not adequately stimulate the IGF-1–Akt pathway to result in skeletal muscle hypertrophy. These data suggest that there may be a role for rigorous acid–base control during exercise to optimize exercise adaptation in CKD.

The effect of exercise on skeletal muscle mitochondria is less clear. In 13 patients with CKD not on dialysis, 12 weeks of resistance exercise resulted in significant increases in mitochondrial DNA copy number¹¹⁴, suggesting an increase in mitochondrial mass. However, another study in 17 patients with CKD reported that 12 weeks of either

combined aerobic and resistance exercise, or aerobic exercise alone, did not result in activation of mitochondrial biogenesis, increase in mitochondrial mass or improvement in $\dot{V}O_{2\text{peak}}$ ¹⁰⁴. A study that compared high-intensity interval training with moderate intensity continuous exercise in 14 patients with CKD not on dialysis also reported no increase in mitochondrial biogenesis or improvement in $\dot{V}O_{2\text{peak}}$ in either group¹¹⁵. Studies of mitochondrial function using ³¹P magnetic resonance spectroscopy have shown impaired skeletal muscle bioenergetics in response to exercise in patients with CKD^{116,117}. Whether this impairment can be improved with exercise training requires further investigation.

Exercise has also been shown to induce antioxidant mechanisms in the general population¹¹⁸. In patients with CKD, unaccustomed aerobic and combined aerobic and resistance exercise resulted in upregulation of NRF2, which activates downstream antioxidant pathways, superoxide dismutase 2 (SOD2), which scavenges oxygen free radicals, and other antioxidant mechanisms; however, this effect was abrogated following exercise training¹⁰⁴. Similarly, 8 weeks of resistance exercise training in people with non-dialysis CKD did not seem to increase skeletal muscle levels of oxidative stress¹⁰⁸, suggesting that exercise training upregulates antioxidant defence mechanisms. Similar findings have been reported in the general population¹¹⁹. These data provide evidence that exercise is safe in terms of effects on oxidative stress and that prolonged periods of training will not result in oxidative damage to DNA, lipids and proteins.

Skeletal muscle contains a population of resident stem cells, known as satellite cells, that are required for cellular repair and regeneration. Upon activation, satellite cells can proliferate and differentiate into myoblasts. Evidence suggests that the process of satellite cell activation following resistance exercise training is impaired in patients with CKD and in those on dialysis. A study in patients with CKD reported that 8 weeks of resistance exercise training did not result in an increase in the post-exercise mRNA expression of either of the two key myogenic regulatory factors MyoD or myogenin, despite significant reductions in myostatin mRNA expression following exercise¹⁰⁸. As MyoD and myogenin signal the differentiation of myoblasts into myotubes¹²⁰, it is likely that the lack of upregulation of this pathway in response to exercise would impair the activation of satellite cells; however, satellite cell activation was not directly assessed in this study. A study in patients on haemodialysis or peritoneal dialysis found an abnormality in satellite cell activation that was confined to type II fibres, with an increase in satellite cells following resistance exercise training in type I fibres¹⁰⁰. This finding is consistent with a more severe catabolic effect of CKD on type II than type I fibres. Importantly, the defect in satellite cell activation in these fibres does not seem to be affected by exercise training¹⁰⁰.

Very little is known about the molecular effect of exercise in KTRs. One study showed that the change in skeletal muscle mRNA expression that occurs in response to acute exercise was the same before and after transplantation and involved activation of pathways that contribute to inflammation and to extracellular matrix and cytoskeletal interactions⁹⁴. This interesting finding suggests that improvement in kidney function owing to transplantation does not affect the response of skeletal muscle to exercise.

In summary, aerobic and resistance exercise have important beneficial effects on skeletal muscle and on mediators of muscle wasting (Fig. 2). However, not all the pathological changes within muscle that occur in CKD seem to be responsive to exercise. More data are required to reach a consensus on the molecular effects of exercise on skeletal

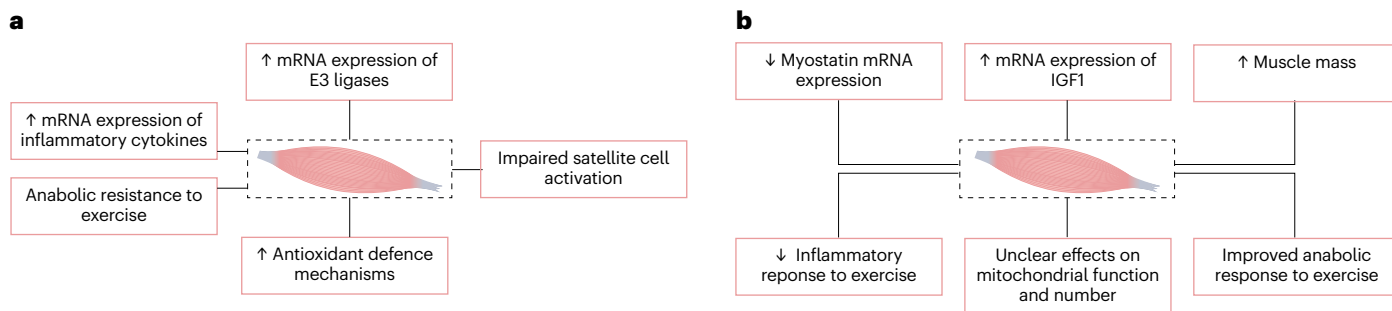


Fig. 2 | Key effects of exercise on skeletal muscle in CKD. **a**, A one-off bout of exercise in an untrained individual with chronic kidney disease (CKD) is associated with increases in the messenger RNA (mRNA) expression of inflammatory cytokines and E3 ligases, increases in antioxidant defence mechanisms, anabolic resistance to the exercise stimulus and impaired satellite

cell activation. **b**, Longer-term effects of exercise training in CKD include decreases in myostatin mRNA expression, reduced inflammatory responses to exercise, improvements in the anabolic response, expression of IGF-1 mRNA and increases in muscle mass. IGF1, Insulin like growth factor-1.

muscle across all stages of CKD and particularly in KTRs who continue to experience muscle wasting and fatigue after transplantation. In the general population, exercise has beneficial effects on maintaining the integrity of the nervous system, including the neuromuscular junction¹²¹, but nothing is known about neuromuscular junction health in human CKD. The effect of exercise on the neuromuscular junction is clinically important because deterioration of this junction is a hallmark and an initiating factor in sarcopenia in the healthy ageing population and in people with chronic disease¹²².

Bone health

The healthy kidney has a key role in maintaining calcium and phosphate homeostasis through tightly regulated mechanisms involving parathyroid hormone (PTH), vitamin D, fibroblast growth factor 23 (FGF23) and klotho¹²³. As kidney function deteriorates these mechanisms are impaired, contributing to abnormalities in bone turnover, mineralization, volume, linear growth or strength as well as vascular or other soft-tissue calcifications. These disturbances, collectively defined as CKD-mineral and bone disorder (MBD)¹²⁴, substantially contribute to the higher incidence of morbidity and mortality in patients with CKD, particularly bone fractures¹²⁵ and CVD¹²⁶.

Higher levels of physical activity are widely accepted to promote bone health across the lifespan by improving bone mass in growing children and decreasing bone loss in older adults¹²⁷. Physical activity therefore reduces the risk of osteoporosis and bone fractures¹²⁸. However, not all physical activities are equally effective and only those that involve high-impact multi-directional movement patterns and/or unaccustomed loading are thought to provide optimal osteogenic stimulus¹²⁹. These activities include weight-bearing impact loading exercises (such as hopping and jumping) and progressive resistance (muscular strength) exercise training. Preliminary evidence from clinical studies¹³⁰ and animal models¹³¹ suggests that physical exercise might have a protective role in counteracting CKD-MBD.

Bone cell mechanotransduction is a complex process that involves several pathways leading to bone formation, of which the Wnt- β -catenin cascade is one of the most prominent¹³². Activation of this pathway occurs through binding of Wnt ligands to a co-receptor complex consisting of low-density lipoprotein receptor-related protein 5 (LRP5) or LRP6 and frizzled, leading to β -catenin-induced gene transcription that promotes osteoblast differentiation and osteocyte formation.

This pathway is negatively regulated by endogenous secreted inhibitors such as sclerostin (and DKK1) that competitively bind to LRP5 or LRP6. Sclerostin is downregulated by mechanical loading and upregulated by mechanical disuse¹³³.

In CKD, the Wnt- β -catenin pathway is downregulated and serum levels of sclerostin are elevated as kidney function declines¹³⁴. Hence, downregulation of sclerostin is a target to promote bone formation and adaptation. A study in 193 patients on haemodialysis demonstrated that dynamic resistance exercise training (but not isometric resistance exercise training) performed before each dialysis session (i.e. three times per week) for 6 months reduced sclerostin levels and increased bone mineral density¹³⁵. Another pilot study in 13 patients on haemodialysis found no change in sclerostin levels following 8 weeks of intradialytic resistance exercise but reported an increase in bone alkaline phosphate, which is suggestive of increased osteoblast activity¹³⁶. By contrast, a secondary analysis of a 6-month, three times per week, aerobic exercise trial in patients with CKD who were overweight or had obesity did not report any changes in markers of bone metabolism, including sclerostin¹³⁷. This result is perhaps unsurprising considering the low to moderate intensity of the exercise intervention.

The RANKL-RANK-OPG system has an essential role in osteoclast differentiation and bone resorption¹³⁸. Binding of receptor activator of NF- κ B (RANK) ligand (RANKL) to its receptor RANK induces osteoclastogenesis, whereas osteoprotegerin (OPG), a soluble decoy receptor for RANKL, inhibits osteoclastogenesis by preventing RANKL binding to RANK. Notably, binding of RANKL to leucine-rich repeat-containing G protein coupled receptor 4 (LGR4) promotes bone formation but might also promote vascular calcification. In CKD, the RANKL-RANK-OPG system favours bone resorption as PTH upregulates RANKL and downregulates OPG¹³⁹. By contrast, exercise might inhibit bone loss by reducing RANKL and/or increasing OPG¹⁴⁰. A study in 14 patients on haemodialysis reported that 6 months of intradialytic resistance exercise increased plasma OPG levels compared with a control group ($n = 12$) who did not exercise¹⁴¹. An earlier study from the same group showed improved bone mineral density following a similar exercise programme¹³⁶, suggesting that the increase in OPG levels might contribute to prevention of bone loss.

FGF23 is a hormone produced mainly in mature osteoblasts and osteocytes that has a key role in regulating bone and mineral metabolism. The physiological actions of FGF23 include promotion

of phosphaturia and inhibition of PTH, which are also dependent on the presence of FGF receptors and the co-receptor klotho. As kidney function declines, klotho expression is reduced and generates a state of FGF23 resistance¹⁴². Hence, patients with CKD have reduced levels of klotho¹⁴³ and increased levels of FGF23 and PTH¹⁴⁴. Dynamic (but not isometric) resistance exercise training increased klotho levels and reduced levels of FGF23 and PTH in patients on haemodialysis¹³⁵. Moreover, 6 months of resistance exercise with or without blood flow restriction training increased klotho levels and reduced FGF23 in patients with stage 2 CKD¹⁴⁵. By contrast, a substudy of the RENEXC RCT found no effect of 12 months of either strength or balance training in combination with aerobic exercise on FGF23 levels in 151 patients with stage 3–5 CKD who were not on dialysis⁸⁷. This conflicting result is hard to interpret as no control group was included in the substudy and both interventions had similar effects on physical performance.

In addition to their mechanical interaction, muscle and bone crosstalk and regulate each other through the production of myokines and osteokines¹⁴⁶. For example, muscle-derived IL-6 and myostatin promote bone resorption by inducing osteoclastogenesis largely via RANKL stimulation. Myostatin also increases the production of sclerostin and DKK1, which leads to reduced osteoblastogenesis and bone formation¹⁴⁷. As discussed above, IL-6 and myostatin are elevated in CKD but this increase can be abrogated through exercise^{11,101,108,110}, which could also have a beneficial effect on bone metabolism. This hypothesis is supported by evidence that the favourable effects of resistance exercise training (with or without blood flow restriction) on the klotho-FGF23 axis noted above were accompanied by a reduction in IL-6 (and other favourable effects on the inflammatory profile)¹⁴⁵.

Osteokines have been implicated in exercise-induced improvements in muscle metabolism and consequently might contribute

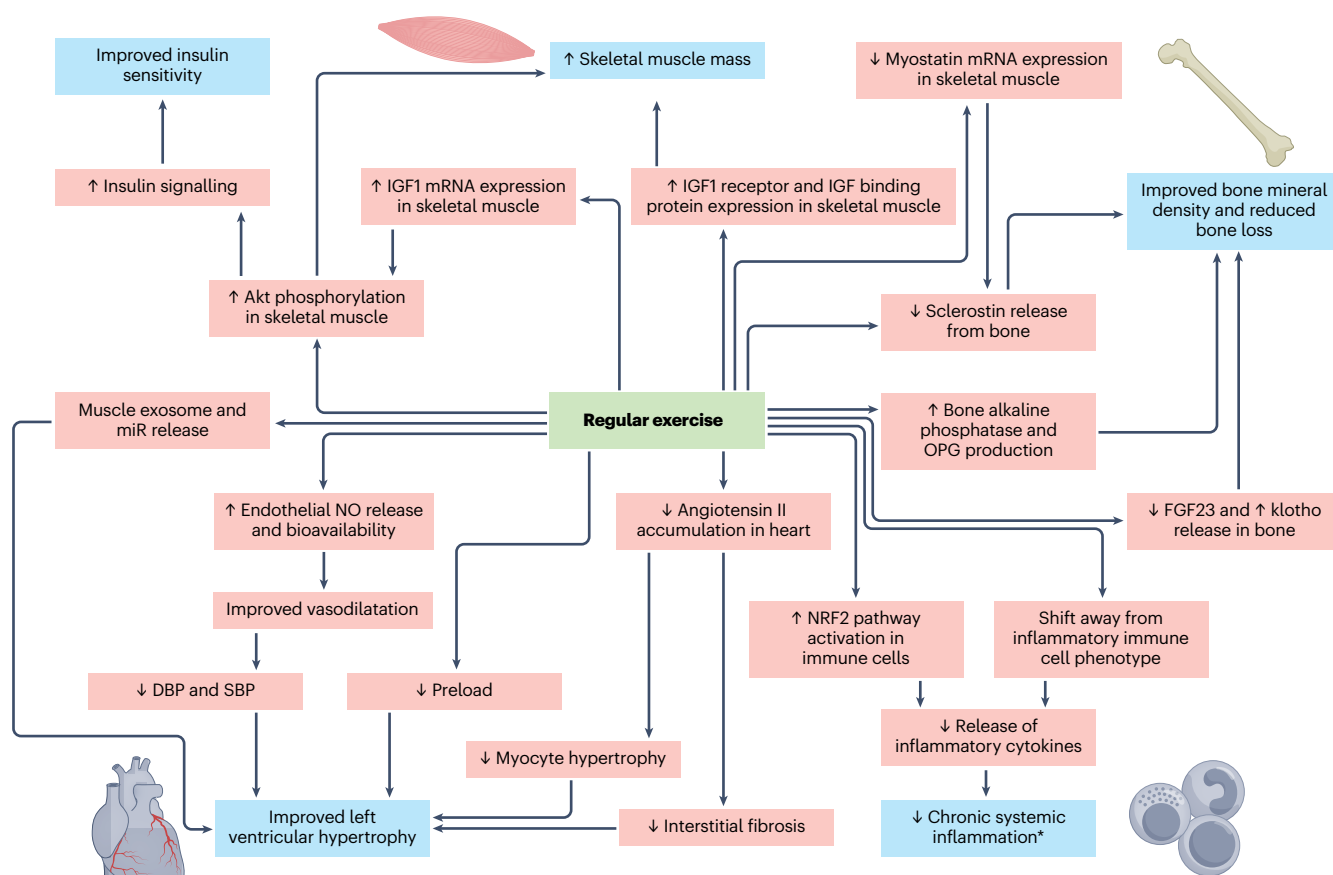


Fig. 3 | Potential mechanisms that could underlie the physiological benefits of aerobic and resistance exercise in CKD. Regular exercise promotes a shift towards a less inflammatory immune cell phenotype and reduces chronic systemic inflammation, which likely contributes to improvements in left ventricular hypertrophy, insulin sensitivity, skeletal muscle mass and bone mineral density. Left ventricular hypertrophy is one of the hallmarks of uraemic cardiomyopathy. Exercise might have beneficial effects on left ventricular hypertrophy via improvements in afterload (e.g. improved vasodilatation and reduced blood pressure) and preload factors, myocyte hypertrophy and interstitial fibrosis as well as via the release of muscle exosomes and microRNA (miR). Exercise-mediated increases in Akt phosphorylation in skeletal muscle via a direct action and/or via increased insulin-like growth factor 1 (IGF1) expression may contribute to improved insulin signalling and sensitivity as well as increased

skeletal muscle mass. Exercise-related improvements in bone mineral density and reduced bone loss may be mediated via a decrease in sclerostin release from bone either directly and/or via a reduction in skeletal muscle myostatin mRNA expression. Exercise-mediated decreases in the release of fibroblast growth factor-23 (FGF23) and increases in klotho release in bone might also contribute to improved bone mineral density and reduced bone loss. These potential mechanisms are based on the available data from studies of the effects of short-term exercise intervention in patients with chronic kidney disease (CKD), kidney failure on dialysis and kidney transplant recipients. The available data are limited, and larger, longer duration intervention studies are needed to confirm these potential mechanisms across the spectrum of CKD. Akt, protein kinase B; DBP, diastolic blood pressure; NO, nitric oxide; NRF2, nuclear factor erythroid-2 related factor 2; OPG, osteoprotegerin; SBP, systolic blood pressure.

to the beneficial effects of exercise on sarcopenia. Osteocalcin can positively regulate muscle mass and function, whereas sclerostin and RANKL have been linked with muscle catabolic pathways^{146,147}. Studies in people with CKD have failed to detect an exercise-induced increase in osteocalcin^{137,141}, but the available data are extremely limited and contrast with the favourable effects of exercise on sclerostin and the RANKL system. Moreover, evidence from animal studies indicates that exercise-induced increases in bone-derived osteocalcin and muscle-derived IL-6 are dependent on each other^{148,149}. Given the roles of the Wnt- β -catenin, RANKL-RANK-OPG and FGF23-klotho pathways in promoting vascular calcification in CKD^{139,142}, it is possible that the promising effects of exercise on bone health might also help to reduce the risk of CVD in people with kidney disease.

In summary, physical exercise promotes bone health and might counteract CKD-MBD, but evidence in CKD populations is limited. Early studies in patients on haemodialysis suggest that exercise might activate the canonical wnt pathway leading to bone formation and inhibit bone resorption by upregulating OPG^{135,141}. The ability of exercise to downregulate RANKL should also be investigated in CKD. Preliminary evidence in patients with early-stage CKD and in those on haemodialysis suggest that exercise can help to regulate abnormal bone-derived hormone release by increasing klotho and reducing FGF23 (refs. ^{135,145}). Further research is needed to increase our understanding of the effect of exercise in people with CKD on the release of myokines that regulate bone metabolism and the release of osteokines implicated in muscular and cardiovascular health.

Conclusions

Compared with the volume of studies describing the physiological benefits of regular exercise for people living with CKD, investigations into the mechanisms that underlie these benefits are limited in number and most have focused on patients receiving haemodialysis. Many mechanisms by which regular exercise confers physiological benefits in other populations require confirmation in CKD and larger studies with longer durations of interventions (≥ 12 weeks) are needed. The benefits of exercise participation are likely to vary between people with various stages of CKD, kidney failure on dialysis and KTRs, not least because physical ability and the impact of kidney disease on physiological outcomes will differ between stages of kidney disease and between individuals. However, the available data suggest that the mechanisms by which exercise results in improvements in cardiovascular outcomes, insulin resistance and muscle and bone health in CKD are multifaceted, involving locally released substances and crosstalk between tissues, particularly via muscle-derived mediators (Fig. 3). An overarching mechanism that seems to have a role in many of the beneficial effects of exercise in CKD is a reduction in chronic systemic inflammation with reductions in the inflammatory characteristics of immune cells and cytokine release. Further research is needed to investigate whether exercise-mediated reductions in chronic systemic inflammation also have beneficial effects on other comorbidities of CKD with inflammatory aetiology such as the development of cerebral atrophy and cognitive impairment. Improved understanding of the mechanisms that underlie the physiological benefits of exercise in CKD, kidney failure and kidney transplantation could lead to the development of evidence-based targeted exercise guidance to provide optimal benefit for individual patients and potentially to the use of exercise interventions as an adjuvant therapy to pharmaceutical treatments.

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Additional information

Correspondence should be addressed to Nicolette C. Bishop.

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¹School of Sport, Exercise and Health Sciences and National Centre for Sport and Exercise Medicine, Loughborough University, Loughborough, UK.

²National Institute for Health Research (NIHR) Leicester Biomedical Research Centre, University Hospitals of Leicester NHS Trust and University of Leicester, Leicester, UK. ³Department of Cardiovascular Sciences, University of Leicester, Leicester, UK. ⁴John Walls Renal Unit, University Hospitals of Leicester NHS Trust, Leicester, UK. ⁵Faculty of Sport Sciences, Waseda University, Tokorozawa, Japan. ⁶Department of Sports Science and Physical Education, The Chinese University of Hong Kong, Hong Kong, China. ⁷Research Centre in Sports Sciences, Health Sciences and Human Development, University of Maia, Maia, Portugal.